EX. No. 2.1: CREATE A 3D SOLID MODEL FROM ISOMETRIC DRAWING

Date: _____

AIM

To design and develop a 3D solid model from the given isometric drawing using Autodesk Fusion 360.

REQUIREMENTS

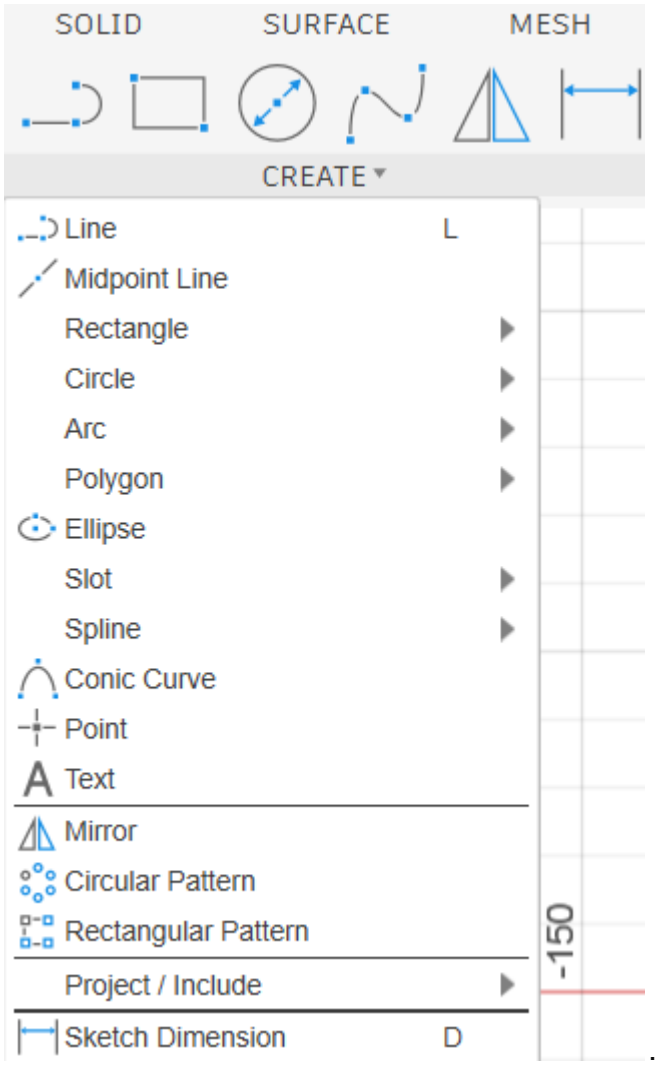
- Autodesk Fusion 360 software
- Windows 10 operating system
- 16 GB RAM recommended
- Internet connection

PROCEDURE

Step 1: Open Fusion 360

Launch Autodesk Fusion 360 and create a new design using the New Design option.

Step 2: Create a New Sketch

Select the Sketch tool and choose the Front plane to start drawing the base profile 

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Step 3: Draw the Base Profile

Use the Rectangle tool to draw a rectangle with dimensions of 100 mm × 60 mm. Apply the required dimensions and click Finish Sketch.

Step 4: Create the Base Solid

Select the rectangular profile and use the Extrude command. Set the extrusion distance to 30 mm and click OK.

Step 5: Sketch on the Solid Face

Select the required face of the solid and create a new sketch to add the next feature.

Step 6: Draw the Circular Profile

Choose the Circle tool and draw a circle with a diameter of 100 mm at the position indicated in the isometric drawing. Finish the sketch.

Step 7: Add the Circular Feature

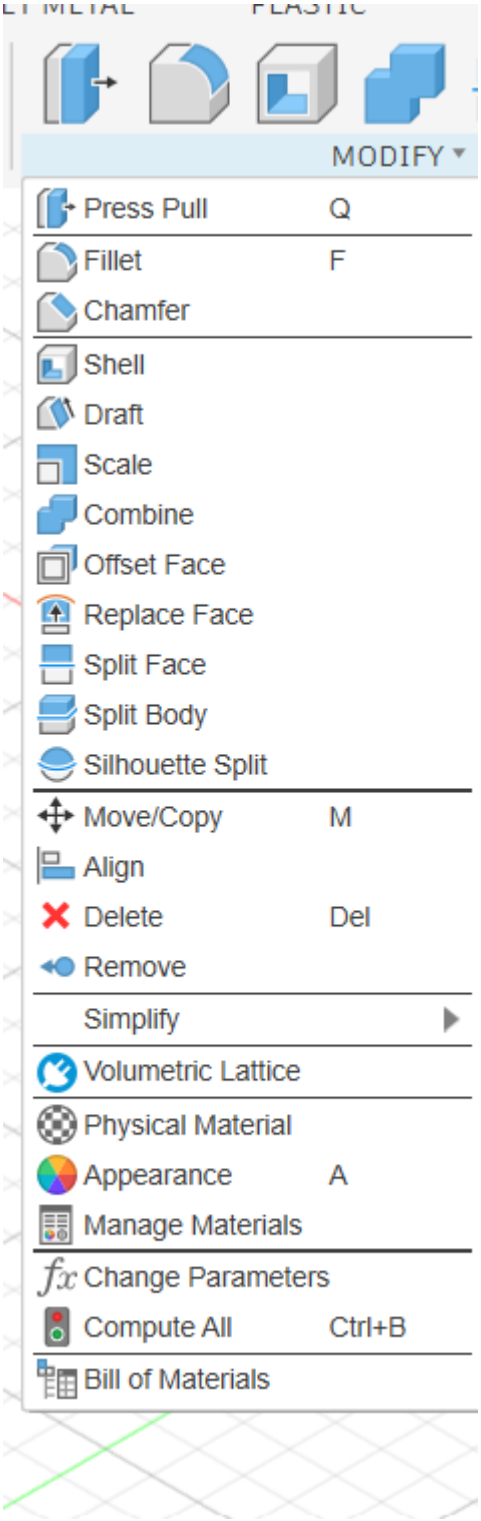
Select the circular profile and use the Extrude command with a distance of 20 mm. Choose the appropriate operation based on the required shape.

Step 8: Create Additional Features

Select the suitable face of the model and start a new sketch. Use the Line and Rectangle tools to draw the additional profiles shown in the isometric drawing.

Step 9: Modify the Model

Apply the required dimensions and constraints to the sketch. Use the Trim tool



wherever necessary to remove unwanted lines.

Step 10: Extrude the Additional Profiles

Finish the sketch and extrude the closed profiles to the required height or depth according to the given drawing.

Step 11: Complete the Solid Model

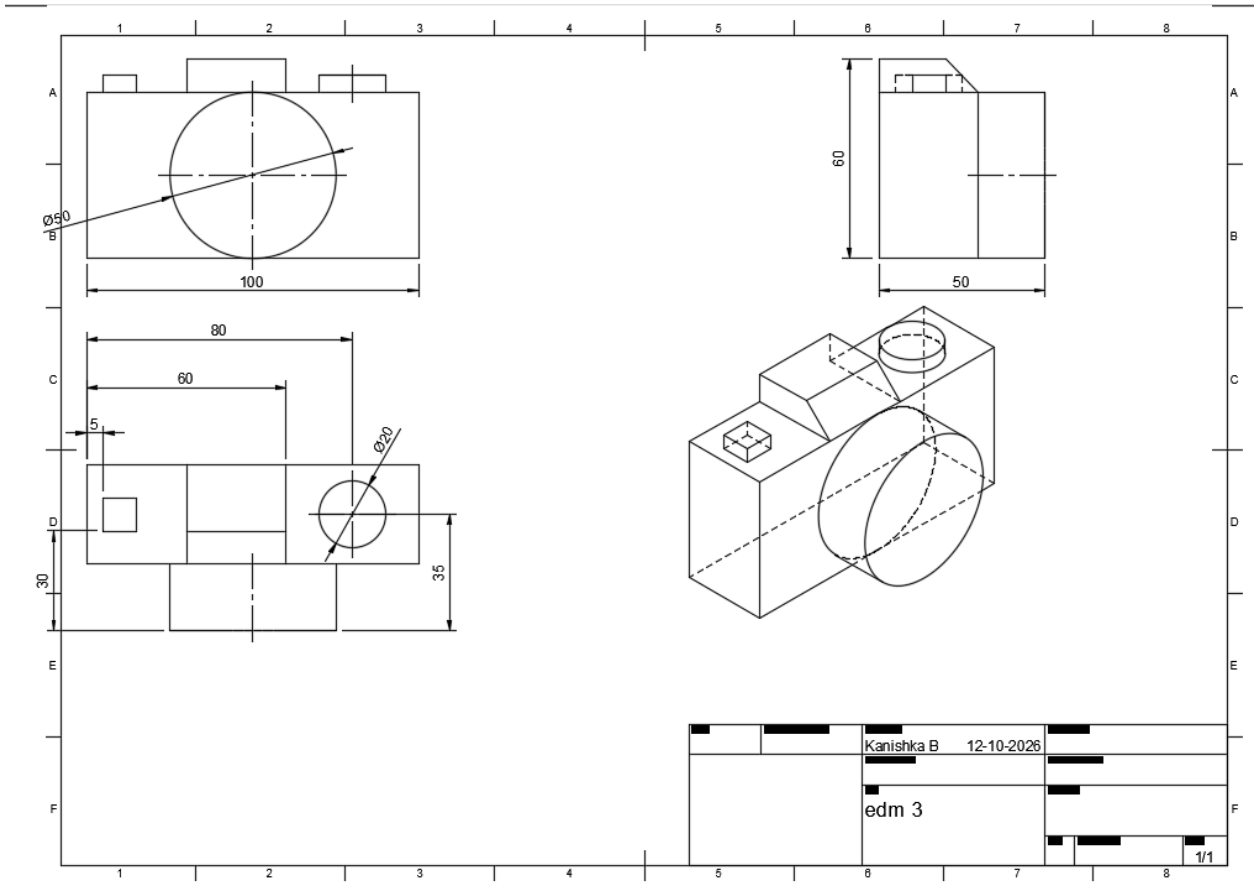
Use the Fillet or Chamfer command if required to finish the edges. Check the model from different viewing angles and save the design.

INPUT

Given isometric drawing.

OUTPUT

3D solid model created using Autodesk Fusion 360.



RESULT

Thus, the 3D solid model was successfully created from the given isometric drawing using Autodesk Fusion 360.

